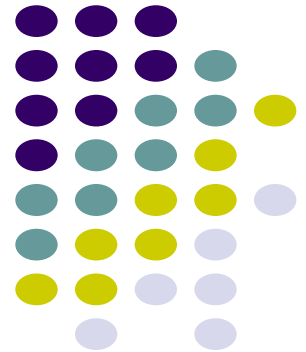


Thrashing and Storage Devices

ECE 469, April 07

Aravind Machiry



RECAP: Page Replacement Algorithms



- Optimal
- FIFO
- Random
- Approximate LRU (NRU)
- FIFO with 2nd chance
- Clock: a simple FIFO with 2nd chance
- Enhanced FIFO with 2nd chance

Whose pages should be replaced?



- Global replacement:
 - All pages from all processes are lumped into a single replacement pool
 - Most flexibility, least (performance) isolation
- Local replacement
 - Per-process replacement:
 - Each process has a separate pool of pages
 - Per-user replacement:
 - Lump all processes for a given user into a single pool
- In local replacement, must have a mechanism for (slowly) changing the allocations to each pool

Why we need paging? Handling low memory



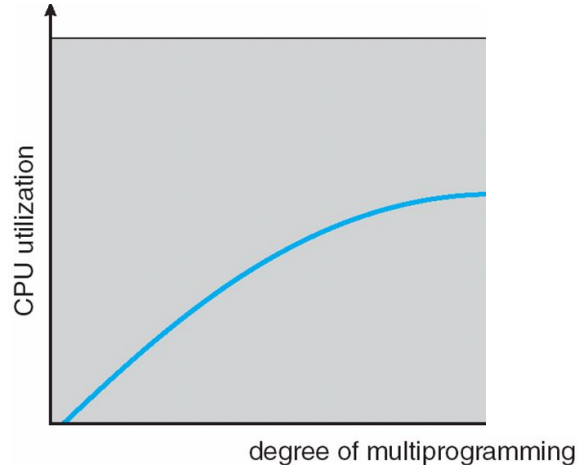
- Suppose you have 8GB of main memory
- Can you run a program that its program size is 16GB?
 - Yes, you can load them part by part
 - This is because we do not use all of data at the same time
- Can your OS do this execution seamlessly to your application?

Why we need paging? Efficient use of memory!



- Process exhibit locality - not all pages of a process need to be in memory!
- Bringing in only required pages allows us to execute multiple processes seamlessly:
 - Increases CPU utilization

Increasing multiprogramming increases CPU utilization!!?



What happens when there is not enough physical memory?



- Suppose **many** processes are making frequent references to 50 pages, memory has 49
- Assuming LRU
 - Each time one page is brought in, another page, whose content will soon be referenced, is thrown out
- What is the average memory access time?
- The system is spending most of its time paging!
- The progress of programs makes it look like “*memory access is as slow as disk*”, rather than “*disk being as fast as memory*”

Thrashing!!

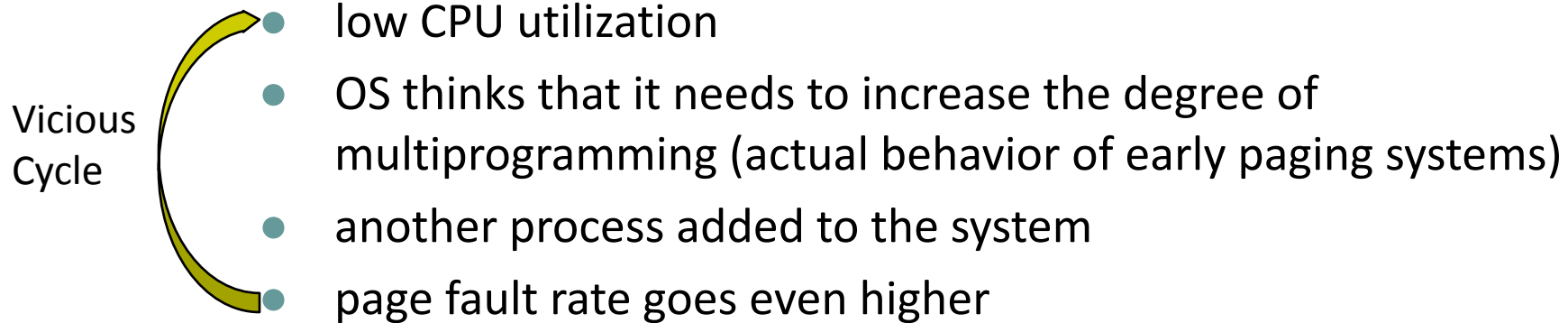
- **Thrashing** $\equiv$ a process is busy swapping pages in and out



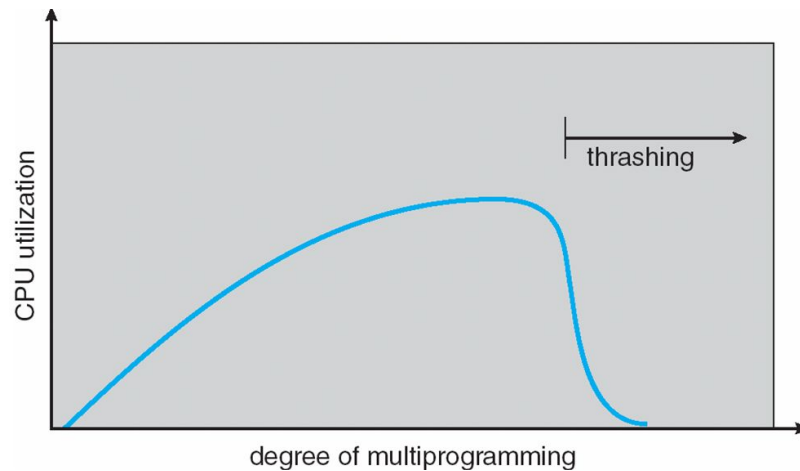


Thrashing can lead to vicious cycle

- If a process does not have “enough” pages, the page-fault rate is very high. This leads to:



Thrashing!!





What causes Thrashing!?

- The system does not know it has taken more work than it can handle
- Virtual memory bites back!
- Mitigating Thrashing:
 - Run fewer programs.
 - Dropping or degrading a course if taking too many than you can handle 😊



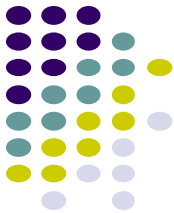
Demand Paging and Thrashing!?

- Why does demand paging work?
 - Data reference exhibits locality
- Why does thrashing occur?
 - Σ size of locality $>$ total memory size



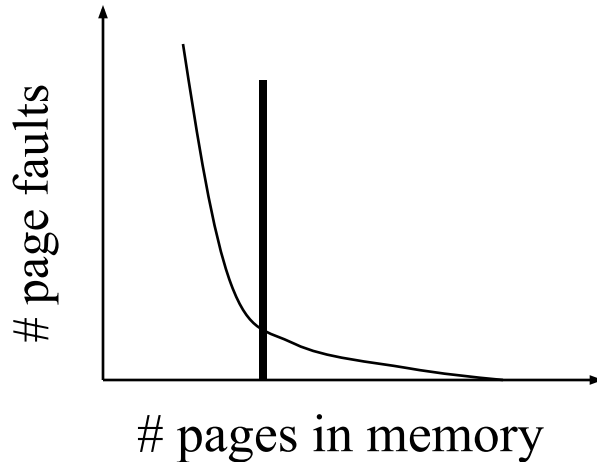
Intuitively, what to do about thrashing?

- If a single process's locality too large for memory, what can OS do?
 - e.g., pin most data (hotter data) in memory, sacrifice the rest
- If the problem arises from the sum of several processes?
 - Figure out how much memory each process needs – “locality”
 - What can we do?
 - Can limit effects of thrashing using local replacement
 - Or, bring a process' working set before running it
 - Or, wait till there is enough memory for a process's need



Key Observation

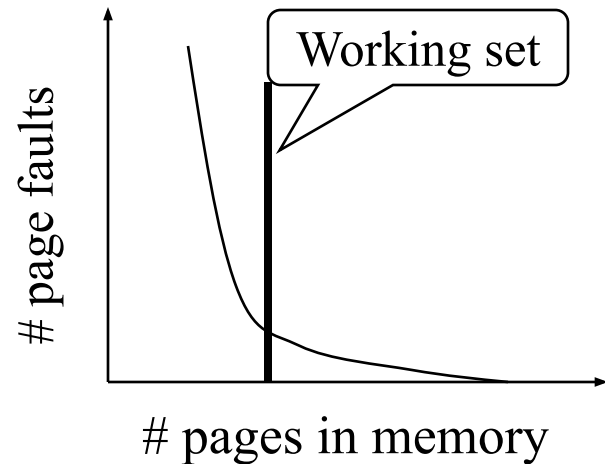
- Locality in memory references
 - Spatial and temporal
- Want to keep a set of pages in memory that would avoid a lot of page faults
 - “Hot” pages
- Can we formalize it?



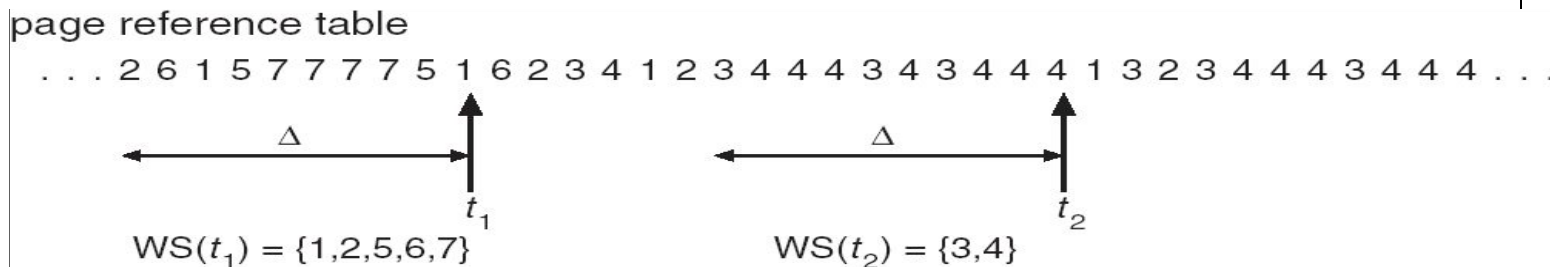
Working Set Model – by Peter Denning (Purdue CS head, 79-83)



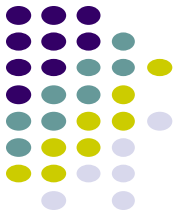
- An informal definition:
 - Working set: The collection of pages that a process is working within a time interval, and which must thus be resident if the process is to avoid thrashing
- But how to turn the concept/theory into practical solutions?
 1. Capture the working set
 2. Influence the scheduler or replacement algorithm



Working Sets

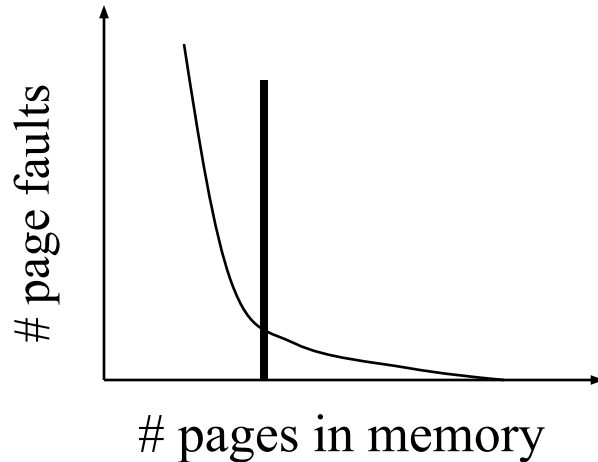


- The working set size is *num of* pages in the working set
 - the number of pages touched in the interval $[t-\Delta+1..t]$.
- The working set size changes with program locality.
 - during periods of poor locality, you reference more pages.
 - Within that period of time, you will have a larger working set size.
- Goal: keep WS for each process in memory.



Working Set Model

- Usage idea: use recent needs of a process to predict its future needs
 - Choose Δ , the WS parameter
 - At any given time, all pages referenced by a process in its last Δ seconds comprise its working set
 - Don't execute a process unless there is enough memory to fit its working set
- Needs a companion replacement algorithm





Working Set Replacement Algorithm

- Main idea
 - *Take advantage of reference bits*
 - *Variation of FIFO with 2nd chance*
- An algorithm (assume reference bit)
 - On a page fault, scan through all pages of the process
 - If the reference bit is 1, clear the bit, **record the current time for the page**
 - If the reference bit is 0, **check the “last use time”**
 - **If the page has not been used within Δ , replace the page**
 - **Otherwise, go to the next page**

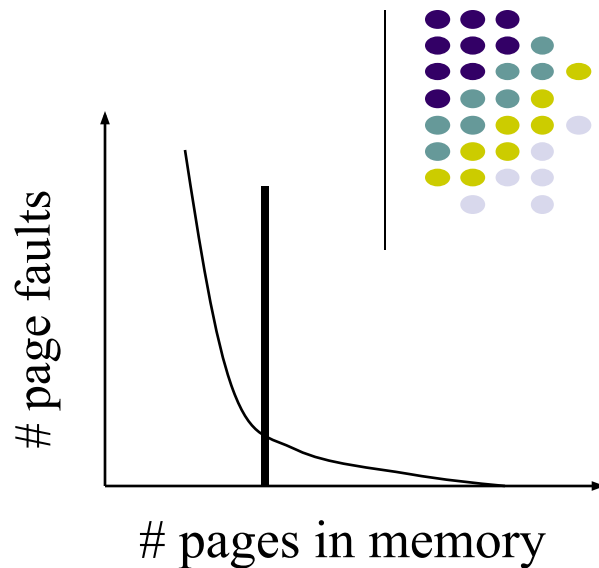
Working Set Clock Algorithm (assume reference bit + modified bit)



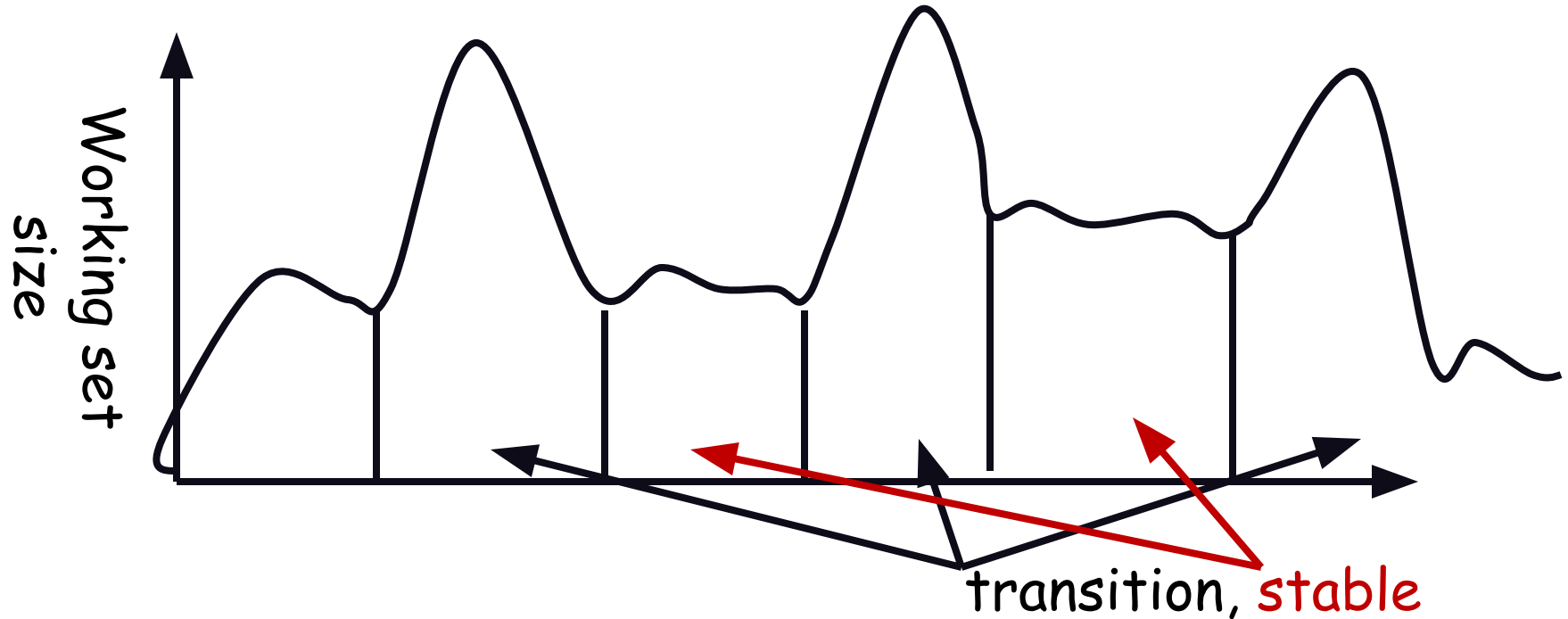
- Upon page fault, follow the clock hand
- If the reference bit is 1, set reference bit to 0, set the current time for the page and go to the next
- If the reference bit is 0, check “last use time”
 - If page used within Δ , go to the next
 - If page not used within Δ and modify bit is 1
 - Schedule the page for page out (then reset modify bit) and go to the next
 - If page not used within Δ and modified bit is 0
 - Replace this page

Challenges with WS algorithm implementation

- What should Δ be?
 - What if it is too large?
 - What if it is too small?
- How many jobs need to be scheduled in order to keep CPU busy?
 - Too few $\square$ cannot keep CPU busy if all doing I/O
 - Too many $\square$ their WS may exceed memory



Working Sets in Real World



More Challenges with WS algorithm implementation



- Working set isn't static
- There often isn't a single "working set"
 - e.g., Multiple plateaus in previous curve (L1 \$, L2 \$, etc)
 - Program coding style affects working set
 - e.g., matrix multiply
- Working set is often hard to measure
 - What's the working set of an interactive program?
 - How to calculate WS if pages are shared?

Storage Devices

- Devices used to store data



Storage Technologies

- Tapes
- Magnetic Disks
- Flash Memory



Tapes

- Low-cost, highly-reliable storage.
- Slow access: ~30MB Per Second.

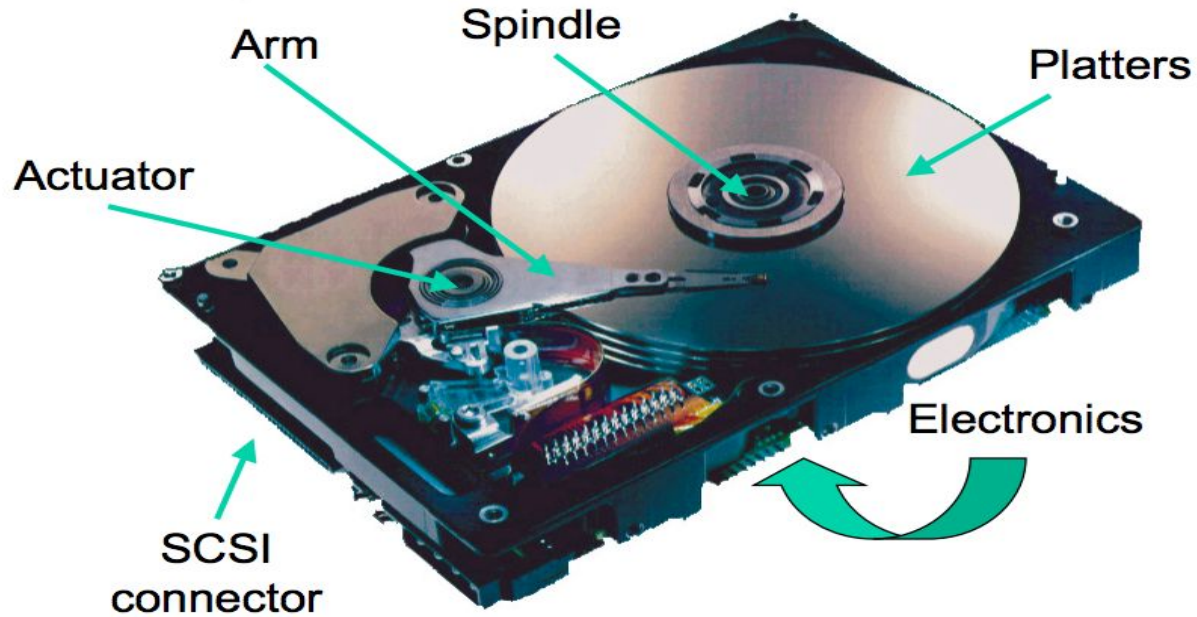


Magnetic Disks

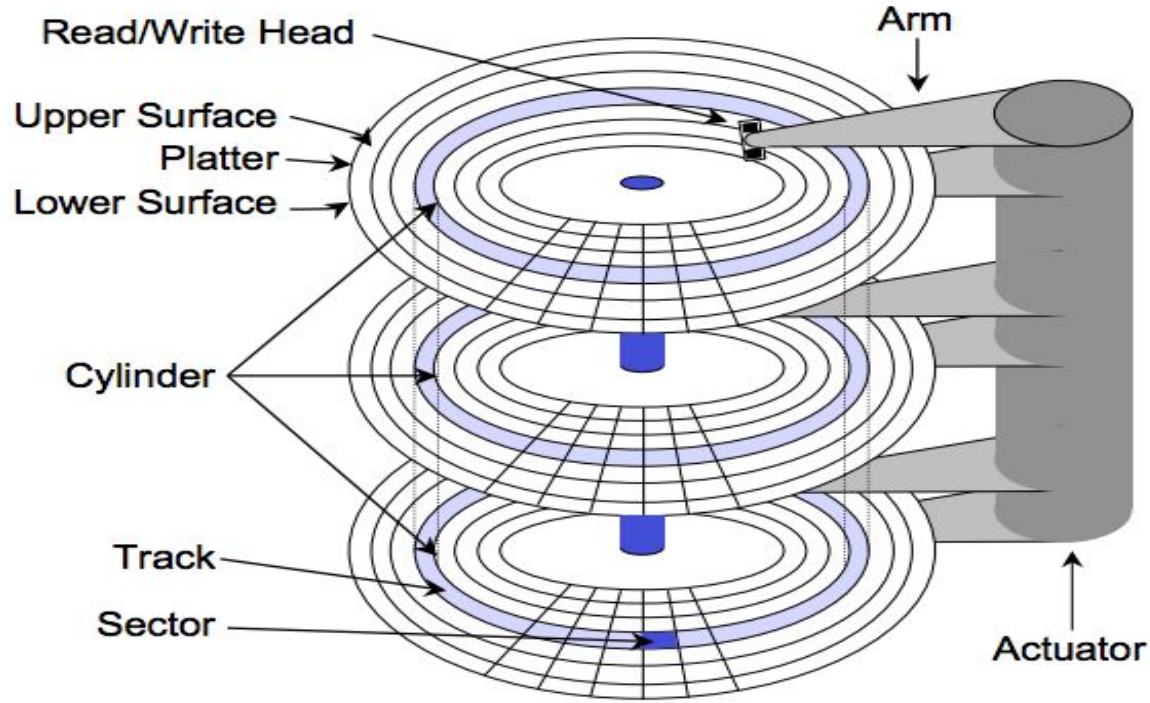


- (Used to be) De Facto standard for storage.
- Medium access: ~150MB Per Second.
- Relatively high failure rate (vs Tape).

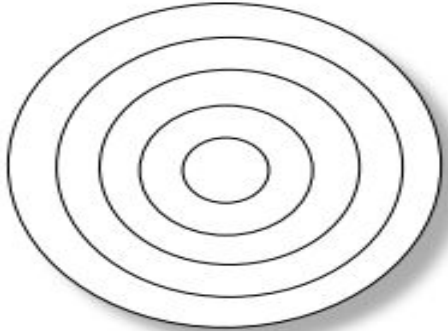
Magnetic Disk



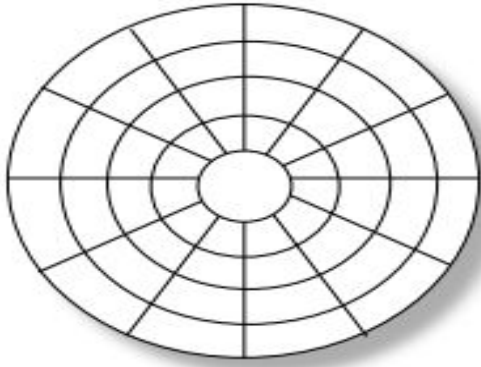
Disk Components



Surface Organized into Tracks

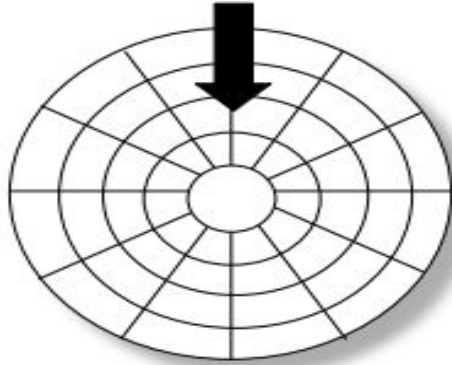


Tracks Broken into Sectors

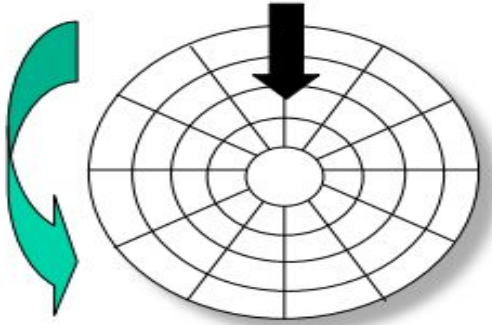


- Disk accesses in the granularity of a sector (usually 512KB)
- This I/O interface is called **block I/O interface**

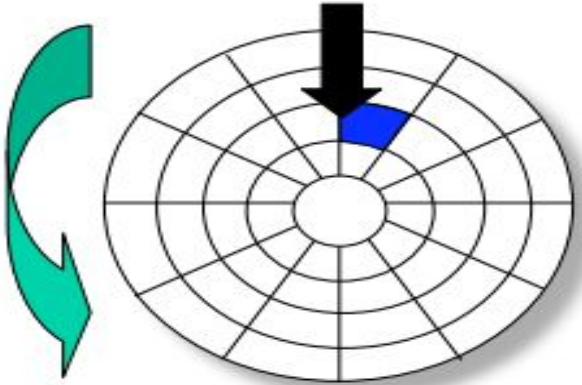
Disk Head Position



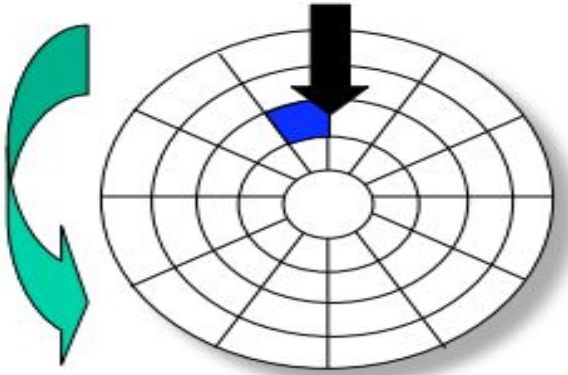
Rotation is Counterclockwise



About to Read Blue Sector

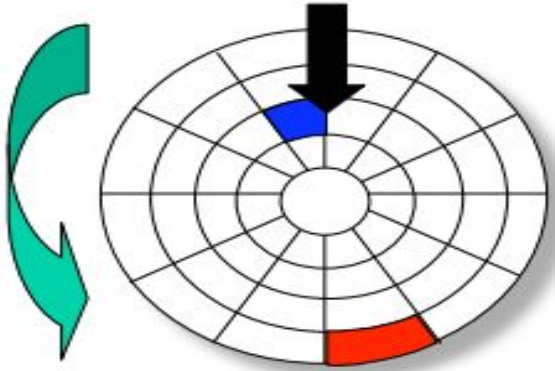


After Reading Blue Sector



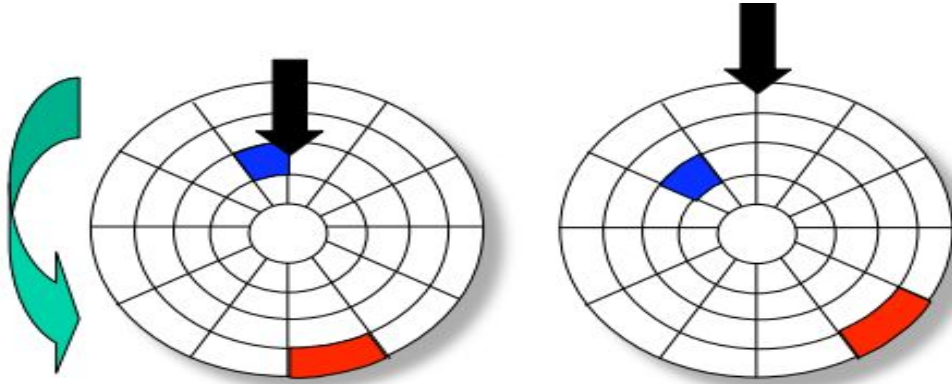
After **BLUE** read

Red Request Scheduled Next



After **BLUE** read

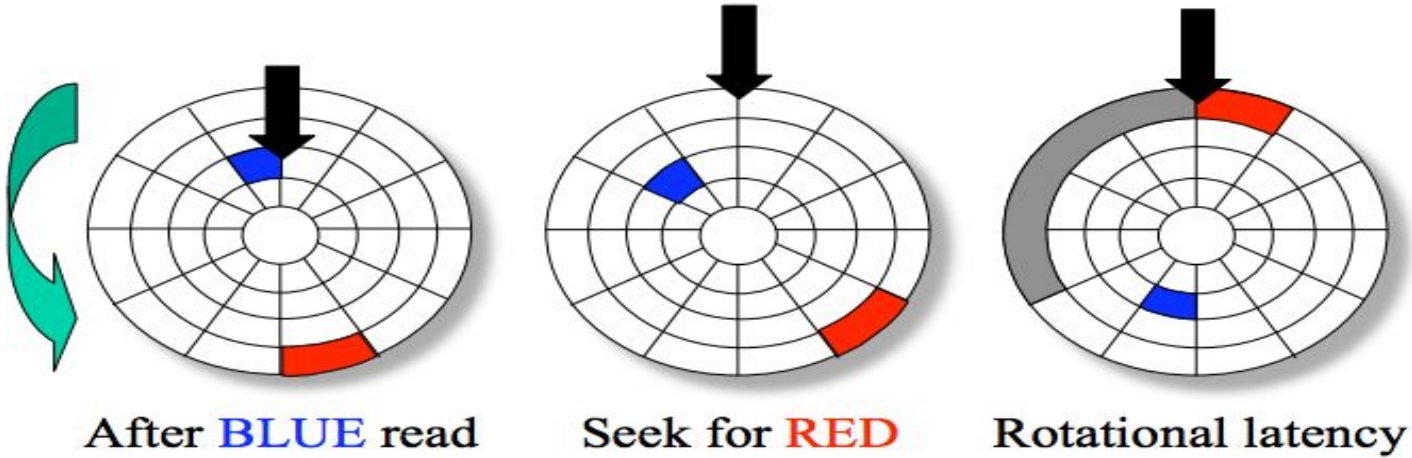
Seek to Red's Track



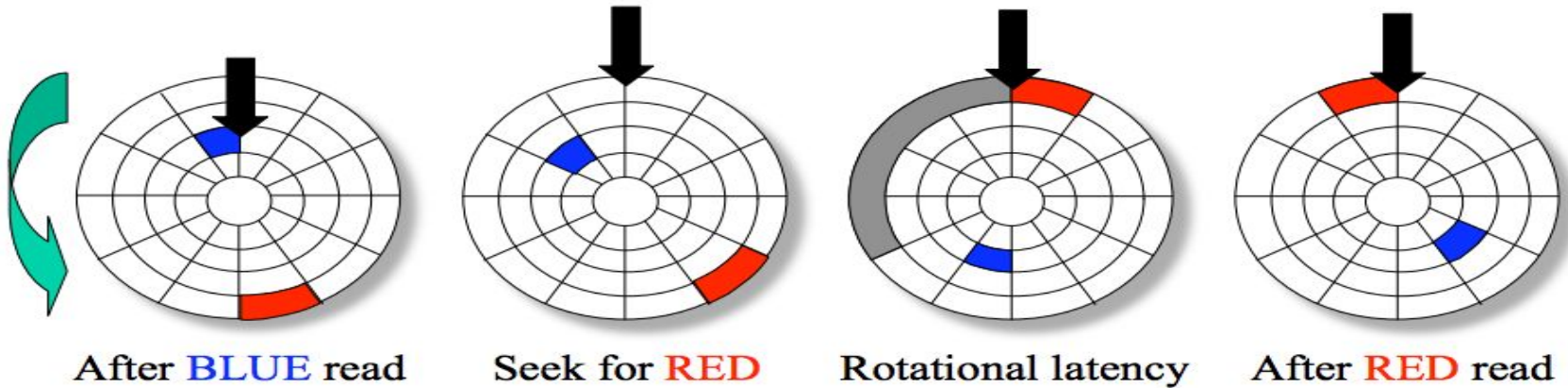
After **BLUE** read

Seek for **RED**

Seek to Red Sector to Reach Head



Read Red Sector



Real numbers for Modern Disks



- # of platters: 1-8
 - 2-16 surfaces for data
- # of tracks per surface: 10s of 1000s
 - same thing as # of cylinders
- # sectors per track: 200-1000
 - so, 100-500KB
- # of bytes per sector: usually 512
 - can be chosen by OS for some disks

Response Time for Disks



- Access time: (service time for a disk access)
 - Once command is received, how long it takes to get the data to OS.
 - Seek + Rotation + Transfer
- Response time:
 - Commands may be queued!
 - Queue time + Access time

Seek Time



- Time required to move head over desired track
 - **Physically** moving the head, not electronic => **slow**!
- A seek has up to four components
 - accelerate
 - coast at max velocity
 - decelerate
 - settle onto correct track
- Seek time depends on workload
 - For random workloads, longer seek time

Rotational Latency



- Time required for the first desired sector to reach head
- Depends on rotation speed
 - measured in Rotations Per Minute (RPMs)
- Computing average rotational latency
 - for almost all workloads, we can safely assume that there is an equal likelihood of landing on any sector of the track
 - this gives equal probability of each rotational latency
 - from 0 sectors to N-1 sectors
 - thus, average rotational latency is time for 1/2 revolution
 - e.g., for 7200 RPM
 - one rotation = $60\text{s} / 7200 = 8.33\text{ ms}$
 - average rotational latency = 4.16 ms

Modern Disk Performance Characteristics



- Seek times: 0.5-15ms, depending on distance
 - average 5-6ms
 - improving at 7-10% per year
- Rotation speeds: 5600-15000 RPMs
 - improving at 7-10% per year

Disk failures

- **Disks fail** more often....
 - When continuously powered-on
 - With heavy workloads
 - Under high temperatures



How disks fail?



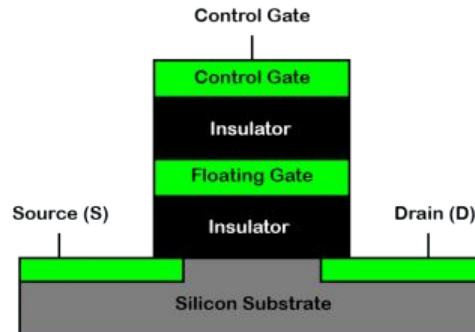
- How do disks fail?
 - Whole disk can stop working (e.g., motor dies, firmware errors)
 - Transient problem (cable disconnected, firmware errors)
 - Individual sectors can fail (e.g., head crash or scratch)
 - Data can be corrupted or block not readable/writable

Fixing disk errors



- Disks can internally fix some sector problems
 - ECC (error correction code): Detect/correct bit flips
 - Retry sector reads and writes: Try 20-30 different offset and timing combinations for heads
 - Remap sectors: Do not use bad sectors in future
 - How does this impact performance contract??

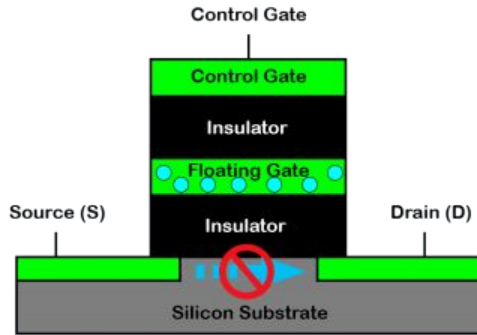
Flash Memory: NAND Cell



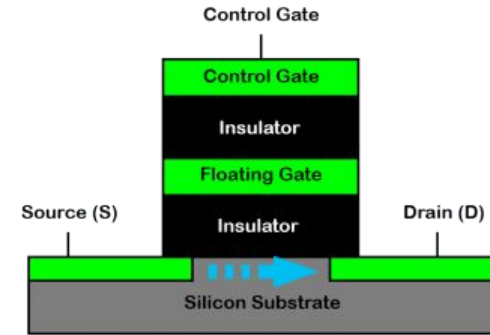
Flash Memory: NAND Cell



Reading



0

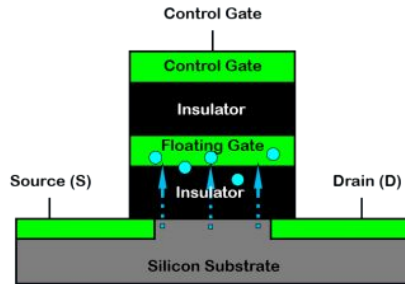


1

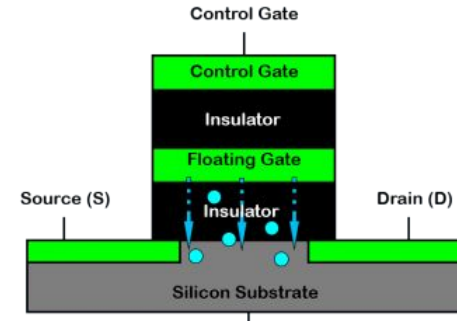
Flash Memory: NAND Cell



Writing



0

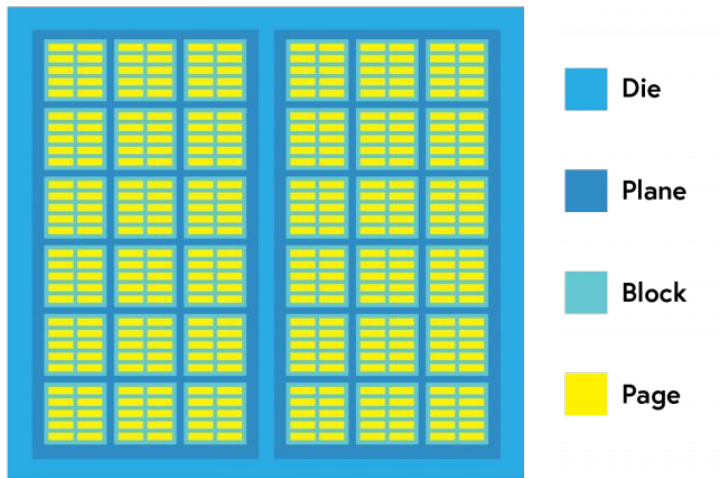


1

Flash Memory



NAND Flash Die Layout



Flash Memory

- Page Size: 512 - 4K bytes.



Write needs complete erasure

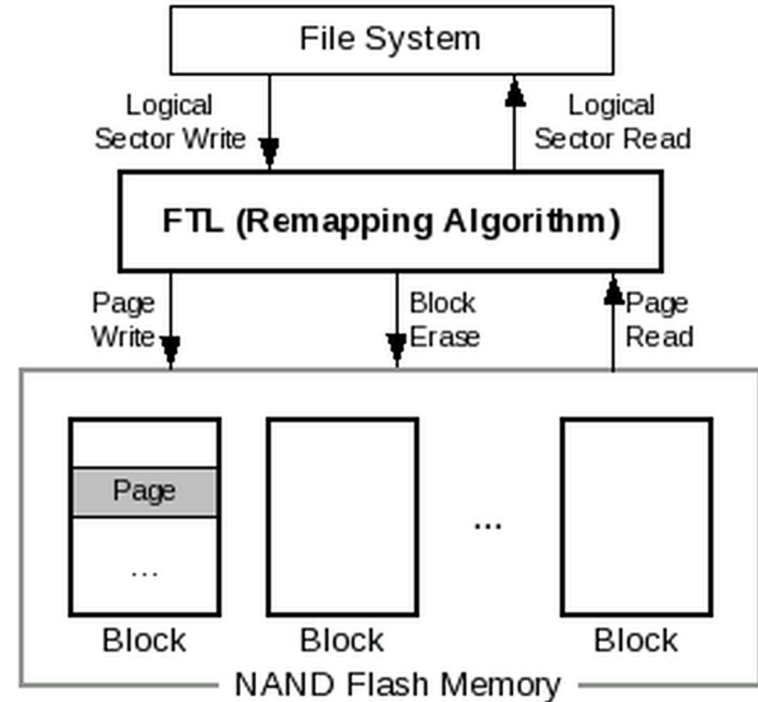


- We cannot set each cell from 0 to 1 (discharging), but can set 1 to 0 (charging)
- To change a 0 to 1:
 - We need to erase entire block to 1
 - And change the required cells to 0
- Writes are **more expensive** than reads.
- Limited number of write cycles --- hardware limitation.

SSD : Flash Translation Layer (FTL)



- Maps logical sectors to real sectors.
- Hides erase before write.
- Maintains free blocks.
- Maintains dirty (invalid) blocks.



FTL



- Mapping from Logical Pages (i.e., sectors) $\leftrightarrow$ Physical Pages
- What do we need?
- What should we do?
 - Read Request?
 - Write Request?

FTL: What we need?



- Translation Table

Logic Page	Physical Page
Page 1	Block 3, Page 4
Page 2	Block 6, Page 7
Page 3	Block 3, Page 5
....	

FTL: Read Request?



- Translation Table

Logic Page	Physical Page
Page 1	Block 3, Page 4
Page 2	Block 6, Page 7
Page 3	Block 3, Page 5
....	

FTL: Read Request?



- Translation Table

Logic Page	Physical Page
Page 1	Block 3, Page 4
Page 2	Block 6, Page 7
Page 3	Block 3, Page 5
....	

- Fetch the corresponding physical page:
 - Page 2 -> return Block 6, Page 7

FTL: Write Request?



- Translation Table

Logic Page	Physical Page
Page 1	Block 3, Page 4
Page 2	Block 6, Page 7
Page 3	Block 3, Page 5
....	

FTL: Write Request?



- Translation Table

Logic Page	Physical Page
Page 1	Block 3, Page 4
Page 2	Block 6, Page 7
Page 3	Block 3, Page 5
....	

- Can we just write to the corresponding physical page?

FTL: Write Request?



- Translation Table

Logic Page	Physical Page
Page 1	Block 3, Page 4
Page 2	Block 6, Page 7
Page 3	(Block 3, Page 5) Block 6, Page 1
....	

Write to Page 3.

- We need to find a free page, i.e., all bits 1
 - **Need Free page list**
 - **e.g., Block 6, Page 1**
- Write the contents to this new page:
 - i.e., Make required bits to 0
- Change the mapping to the new page.

FTL: Write Request?



- Translation Table

Logic Page	Physical Page
Page 1	Block 3, Page 4
Page 2	Block 6, Page 7
Page 3	(Block 3, Page 5) Block 6, Page 1
....	

Write to Page 3.

- What should we do with old page? i.e., Block 3, Page 5

FTL: Write Request?



- Translation Table

Logic Page	Physical Page
Page 1	Block 3, Page 4
Page 2	Block 6, Page 7
Page 3	(Block 3, Page 5) Block 6, Page 1
....	

Write to Page 3.

- What should we do with old page? i.e., Block 3, Page 5
 - Mark it invalid
 - **Need Invalid list**

FTL: Managing Invalid and free list



- Free list:
 - Pages that are not mapped and have all bits 1
- Invalid list:
 - Pages that are not mapped and *does not* have all bit 1

FTL: Managing Invalid and free list



- Free list:
 - Pages that are not mapped and have all bits 1
- Invalid list:
 - Pages that are not mapped and *does not* have all bit 1
- How do pages go from Invalid list to free list? (Garbage collection)

FTL: Managing Invalid and free list



- Free list:
 - Pages that are not mapped and have all bits 1
- Invalid list:
 - Pages that are not mapped and *does not* have all bit 1
- How do pages go from Invalid list to free list?
 - Simple strategy: Wait till all pages in a block become invalid and free the entire block.

FTL: Managing Invalid and free list



- Free list:
 - Pages that are not mapped and have all bits 1
- Invalid list:
 - Pages that are not mapped and *does not* have all bit 1
- How do pages go from Invalid list to free list?
 - Simple strategy: Wait till all pages in a block become invalid and free the entire block.
 - May not be always possible

FTL: Managing Invalid and free list



- Free list:
 - Pages that are not mapped and have all bits 1
- Invalid list:
 - Pages that are not mapped and *does not* have all bit 1
- How do pages go from Invalid list to free list?
 - Find blocks with many invalid pages.
 - Invalidate other pages.(How?)
 - Erase the entire block.

FTL: Managing Invalid and free list



- Free list:
 - Pages that are not mapped and have all bits 1
- Invalid list:
 - Pages that are not mapped and *does not* have all bit 1
- How do pages go from Invalid list to free list?
 - Find blocks with many invalid pages.
 - Invalidate other pages (Move the mapped pages to other free pages).
 - Erase the entire block.

FTL



- Maintains reading performance.
- Amortizes Writing cost:
 - Free list, invalid list
 - Garbage Collection
- Need to have smartness in mapping blocks:
 - Use all pages.
 - Avoid frequently erasing same blocks.